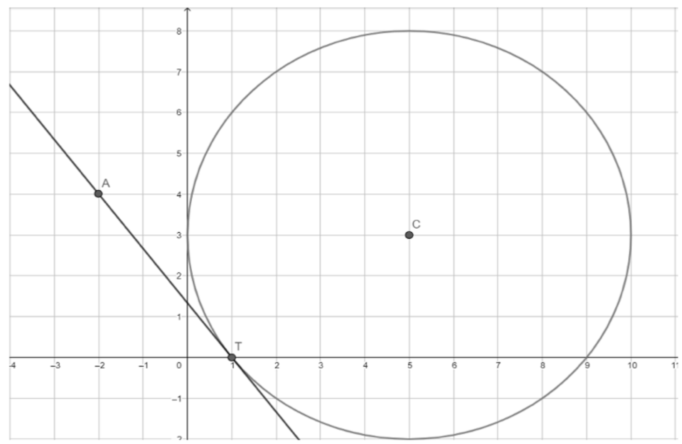
Geometry Name____________________
Unit 13 Date_____________________
Lesson 2 Homework Period____________________

Use the following information for questions 1-4. Circle C is shown, where point T is on the circle and line AT is tangent to Circle C .



1. Use a straightedge to draw a line that goes through points T and C .

2. Write the equation of line AT.

3. Write the equation of line segment CT.

4. Line AT and line CT intersect at point $(1, 0)$. Use the equations of line AT and CT to confirm this point.

Use the following information for questions 5-8. Circle N has a center at $(2, 3)$ and a tangent line with the equation $y = \frac{1}{2}x + 7$.

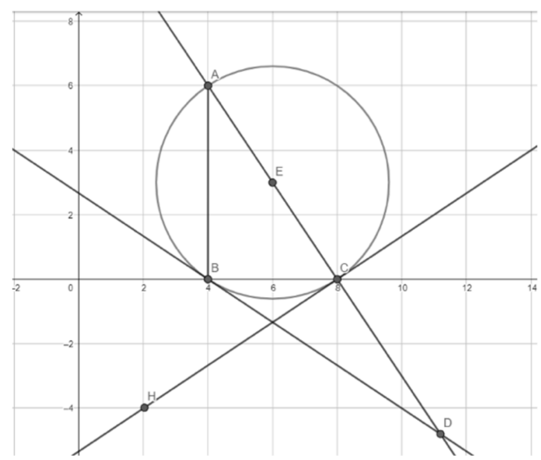
5. Write the equation of the line perpendicular to the tangent line.

6. Find the point where the tangent line touches Circle N. Call this point D.

7. What is the radius of Circle N?

8. Write the equation of Circle N.

Use the following information for questions 9-10. Circle E is shown with secant line AD, tangent lines DB and CH, and chord AB.



9. Complete the table.

Line Segment	Length
$\overline{AB}$	
$\overline{BC}$	
$\overline{AC}$	

10. Write the equation for circle E.